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An Integrated Supplier Selection and Inventory Problem with Multi-sourcing and Lateral Transshipments

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Abstract

In this research, we consider the supplier selection problem of a firm offering a single product via multiple warehouses. The warehouses face stationary, stochastic demand and replenish their inventory via multiple suppliers, to be determined from a set of candidates, with varying price, capacity, quality, and disruption characteristics. Additionally, the warehouses may simultaneously replenish their inventory from other warehouses proactively. With these characteristics, the problem is a multi-sourcing, supplier selection, and inventory problem with lateral transshipments. Even though the benefits of multi-sourcing and lateral transshipments have been presented in the literature individually to mitigate risks associated with uncertain demand and disrupted supply, the intertwined sourcing and inventory decisions under these settings have not been investigated from a quantitative perspective. We develop a decomposition based heuristic algorithm, powered with simulation. While the decomposition based heuristic determines a solution with supplier selection and inventory decisions, the simulation model evaluates the objective function value corresponding to each generated solution. Experimental results show, contrary to the existing literature, inferior decisions may result when considering the selection of suppliers solely on unit and/or contractual costs. We also evaluate the impact of multi-sourcing with rare but long disruptions compared to frequent but short ones.

Keywords: Supplier selection, Inventory optimization, Simulation-optimization, Decomposition, Multi-sourcing, Lateral transshipments

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1. Introduction

As pressure to improve supply chain cost performance increases for many companies, the evaluation and selection of competent suppliers becomes a key concern [25]. In this research, we develop a model to support a broad range of decisions related to strategic sourcing, distribution, and order replenishment of a procurement firm with multiple warehouses. The firm desires to centrally optimize its supplier selection process and inventory replenishment decisions at its warehouses in an integrated fashion. Each warehouse observes stochastic, stationary demand and manages its inventory under a location-specific (nQ, R) policy while incurring a holding and setup cost. Additionally, when the demand at the warehouses cannot be met out of available inventory, a warehouse-specific backorder cost is incurred. We assume that the candidate suppliers, selected centrally, are subject to operational disruptions. We define disruptions with two random variables, corresponding to disruption frequency and disruption downtime. Obviously, disruptions are detrimental in terms of capacity and delivery performance at the suppliers.

An important aspect of this problem is to allow multi-sourcing using both suppliers and warehouses. In other words, a warehouse may source its needs from multiple suppliers or other warehouses proactively. In proactive transshipment strategy, which is the focus in our setting, as opposed to the reactive strategy, transshipments can occur at specific points in time regardless of the demand realization. Shipments from other warehouses are also known as lateral transshipments since the warehouses belong to the same firm. Having multiple suppliers may be more appropriate when (i) suppliers experience capacity issues; (ii) a backup source is required as protection from shortages, strikes, and other supplier disruptions; and (iii) there is a need for maintaining competition among suppliers. Cases (i) and (ii) are the main reasons for the multi-sourcing assumption of this paper.

Despite some clear advantages (e.g., cost and service performance) of multi-sourcing when subject to disrupted and unreliable suppliers, only a limited number of papers present an analytical approach to investigate trade-offs in integrated supplier selection and inventory decision problems (see Section 2). In this paper, we attempt to address this deficiency in the literature. Specifically, our main objective is to develop a cost minimization model for an integrated supplier selection and inventory optimization problem. The cost components include management fees of the selected suppliers; inventory replenishment, holding, and backorder costs; procurement costs of the warehouses; and the total transportation costs between the suppliers and warehouses and/or between the warehouses. To achieve this objective, we determine the (i) selection of suppliers, (ii) assign-

ments of warehouses to selected suppliers, (iii) amount of lateral transshipments among warehouses, and (iv) inventory decisions of each warehouse.

Due to multi-sourcing possibilities for warehouses, this problem poses additional inventory complications. Note that quality (yield), capacity, disruption, lead time, and transportation cost characteristics of a supply source directly influence inventory ordering and timing decisions. Therefore, for various supply sources in our model, we specify a separate inventory policy for each demand and source pair, which is also known as inventory compartmentalization.

Due to uncertainty related to warehouse demand and supplier disruptions, we resort to optimization via simulation [17, 25, 27, 28] to address the complexities of the modeling and analysis of supply chain risk management. We test the performance of our solution approach with/without multi-sourcing and with/without lateral transshipments under various disruption scenarios. We report the influence of specific cost components on supply chain configuration and the computational performance of the proposed solution approach. Specifically, contrary to the existing literature, we demonstrate that an expensive supplier may be preferred to a cheaper supplier depending on the capacity levels and disruption characteristics. We also observe that considerations of multi-sourcing and lateral transshipments in the network result in a lower overall total cost with increased flexibility and robustness. We show that, compared to rare but long disruptions, frequent but short disruptions are less costly. Therefore, supply chain managers should take steps in collaborating with their suppliers in order to shorten their disruptions as possible with a focus on the maintenance activities. However, when a supply chain mostly experiences rare and long disruptions, we show that network designs with multi-sourcing and lateral transshipments provide the largest gains.

In the remainder of the paper, in the next section, we provide a brief literature review. In Section 3, we present the notation, problem description, and formulation. In Section 4, we discuss the details of the solution procedure. The numerical results presented in Section 5 demonstrate the value of the multi-sourcing and lateral transshipments.

2. Relevant Literature

Our work contributes to two main areas: quantitative supplier selection models and multi-supplier inventory models. While the former aims to develop new methodologies leading to an improved supplier selection process, the latter focuses on new and improved replenishment strategies in the presence of multiple suppliers.

The quantitative models for integrated inventory management and supplier selection have been attracting more attention as revealed by the recent review papers [2, 44, 46]. The most common methods include linear programming [4, 32], goal programming [9], mixed integer programming approach [7, 13, 40], a mixed integer nonlinear programming approach [16, 20, 25, 26, 29, 38, 39], and stochastic programming [24, 43].

A similar problem setting with multiple candidate suppliers and multiple warehouses has been considered by Keskin et al. [25, 26], however, without any consideration for multi-sourcing and lateral transshipments. Keskin et al. [26] only consider the trade-off among supplier selection and inventory decisions within a deterministic context and solve the formulation using an efficient generalized bender's decomposition algorithm. Expanding this deterministic model to address stochastic demand and supplier selection under disruptions, Keskin et al. [25] develop a simulation-optimization based solution approach. Their analysis points out that unit costs and management fees influence the selection of suppliers. In many cases, disruptions do not influence the supplier selection decisions, but do impact the inventory decisions. As opposed to Keskin et al. [25], our experimentations show that selecting suppliers solely based on the unit and contractual costs, will lead to inferior solutions. Additionally, due to single sourcing setting, Keskin et al. [25] considers a simpler inventory analysis, without a need for compartmentalization. Most recently, Ventura et al. [48] consider a multi-period multi-stage serial supply chain system with deterministic demand structure in the last stage and all-available suppliers in the first stage. Choudhary and Shankar [14] solve the joint problem of lot sizing, supplier and carrier selection with quantity discount possibilities.

All of these quantitative supplier selection-inventory papers ignore multi-sourcing and lateral transshipment aspects. Even though single sourcing can foster collaboration and partnership, under the existence of capacity shortages, quality issues, and disruptions, relying on a single supplier weakens supply chain robustness and adversely affects supply chain cost performance. Hence, in this paper, we explicitly concentrate on allowing the selection of multiple suppliers, allocating the total demand to the selected suppliers, and determining the (nQ, R) inventory ordering decisions based on the allocated amount. There are other researchers that successfully implemented compartmentalized (Q, R) policies for multi-sourcing without disruptions (see, for instance, Mendoza and Ventura [29]; Mendoza and Ventura [30]; and Pazhani et al. [35]). In fact, both Mendoza and Ventura [30] and Pazhani et al. [35] recommend incorporating supply chain risk (in the form of disruptions), similar to this work, into supplier selection problems. A limited number of papers in

the supplier selection literature [8, 10, 11, 50, 51] concentrate on multi-sourcing. Berger and Zeng [8] present the first analysis comparing single- and multi-sourcing. The relationship between the levels of risk and associated trade-offs is captured by a decision-tree model, and single sourcing is not found to be an effective approach in many situations. Burke et al. [10], given uncertain product demand, simultaneously consider supplier reliabilities, supplier capacities, manufacturer inventory costs, and manufacturer diversification benefits in making integrated supplier selection and inventory allocation decisions. Even with limited experiments, they demonstrate that only when the suppliers have very large capacities, single sourcing is a better option. They extend this paper to consider both upstream (i.e., supply) and downstream (i.e., demand) uncertainty in Burke et al. [11]. Contrary to our results, Burke et al. [11] find that unit cost offered by the supplier (and not the reliability of the supplier) is the key supplier selection criterion. In their paper, Burke et al. [11] consider a profit maximization (as opposed to the cost minimization of this paper) and ignore the fixed cost of supplier selection as well as the impact of quality on total costs. Similar to [10, 11], Zhang and Zhang [51] also consider an integrated supplier selection-inventory problem under stochastic demand with profit maximization. In their model, the inventory problem observed by the buyer is similar to the newsvendor problem. They solve the mixed integer programming formulation with a branch-and-bound based solution approach. In another recent paper, Yu et al. [50] consider a two-stage supply chain in which the buying firm faces a non-stationary, price-sensitive demand of a critical component and that two suppliers (primary and secondary) are available. They determine that either single or dual sourcing can be effective depending on the magnitude of the disruption probability. All of these papers omit the effect of delivery issues related to lead time, varying quality issues at the suppliers, and transportation costs. Our paper presents a unique analysis to estimate the benefit of multi-sourcing and lateral transshipments when suppliers experience disruptions.

A number of papers address inventory problems considering multiple suppliers [18, 19, 22, 30, 33, 37, 41, 42]. Ramasesh et al. [37] show that multi-sourcing provides a greater assurance of timely delivery and greater upside volume flexibility due to diversification of the total requirements at the firm. Parlar and Perry [33] discuss the benefits of multi-sourcing in the presence of supply uncertainty. Federgruen and Yang [19] generalize these works to determine an optimal set of suppliers, the aggregate order, and its allocation to the selected suppliers when suppliers experience disruptions or quality problems. Hajji et al. [22] consider a similar problem with an explicit focus on information sharing control policies. They develop a stochastic dynamic programming formulation

illustrating the structure of the optimal control policy. However, they do not consider availability, capacity, and delivery performances of the suppliers. More recently, Mendoza and Ventura [30] develop models for integrated supplier selection and order quantity allocation with consideration of quality and capacity factors for suppliers, but assuming that suppliers are always available. Pazhani et al. [35] consider a serial inventory system integrating the inventory management with supplier selection decisions and transportation costs. In their analysis, Pazhani et al. [35], discuss the benefits of this integration by comparing their system to a sequential system. Ekici [18] proposes an improvement on the Ghodspour and O'Brien [20] in terms of capacitated suppliers and cyclic ordering policy leading to an easier algorithm and extend their work to account for the exact annual supplier capacity. Ruiz-Torres et al. [41] consider a supply chain network with capability of handling multi-supplier sourcing and supplier disruptions. They consider a flexibility characteristic for each supplier that allows for a contingency shipment in case of disruption in other suppliers. Sawik [42] considers the same setting as Ruiz-Torres et al. [41] excluding the contingency characteristic, and with limited global and local disruption scenarios and provides optimal risk-neutral and risk-averse solutions for both cost and customer service level objective functions. For a more detailed discussion on multi-supplier inventory models, we refer the reader to the review papers by Minner [31] and Tajbakhsh et al. [44]. The main distinction between the previous supplier selection papers related to inventory management and ours is that we do explicitly consider the supplier selection decisions as well as the fixed costs incurred for each selected supplier. We also consider the possibility of lateral transshipments between the warehouses, which as Paterson et al. [34] puts it, has received very little attention in the literature. The only work already investigating the value of transshipments within supplier selection in an implicit manner is Axsater et al. [5]. Their network of N retailers is sourced from a central warehouse where a support/contingency warehouse sources the retailers whenever they encounter shortages in the central warehouse. In our paper, however, we consider a broader sense of lateral transshipment, not only as a contingency shipment but as an alternate source. Majority of the literature in the lateral transshipment area considers the reactive transshipments. In the reactive setting, as the one considered in Axsater et al. [5], a transshipment occurs after the realization of the stochastic demand, and potentially, a stockout occasion. However, another stream of research focuses on the proactive lateral transshipments [1, 3, 6, 12]. Additionally, as mentioned by Paterson et al. [34], regular stock re-balancing via proactive lateral transshipments has benefits for the inventory systems, however little work has been done in proactive lateral transshipments in continuous review settings. Our work is also contributing to fill this gap in the literature.

1 A summary of these previous literature is given in Table 1, with supply chain network design
2 characteristics (sourcing structure, lateral transshipment possibilities, demand structure, and sup-
3 plier features), solution methodology to the problem presented, and finally, the cost components
4 considered as part of the objective function.

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Supply Chain Network Details

Cost Components

Paper	Model	Source	LT	Horizon	Demand	Lead	Supplier			Solution	CC	PC	SC	HC	BC	Tr - C
							Time	DF	DD	Cap	Qual	Selection				
Keskin et al. [25]	MINLP	SS	No	IH	S	S	S	S	S	Yes	Yes	Yes	✓	✓	✓	✓
Keskin et al. [26]	MINLP	SS	No	IH	D	N/A	N/A	N/A	N/A	Yes	No	Yes	✓	✓	✓	✓
Burke et al. [10]	MIP	MS	No	SP	S	N/A	N/A	N/A	N/A	Yes	Yes	Yes	✓	✓	✓	✓
Rosenthal et al. [40]	MIP	MS	No	IH	D	N/A	N/A	N/A	N/A	Yes	Yes	Yes	✓	✓	✓	✓
Ghodsypour & O'Brien [20]	MINLP	MS	No	SP	D	N/A	N/A	N/A	N/A	Yes	Yes	Yes	✓	✓	✓	✓
Basnet & Leung [7]	MIP	MS	No	MP	D	N/A	N/A	N/A	N/A	No	No	Yes	✓	✓	✓	✓
Ding et al. [17]	MIP	MS	No	IH	S	N/A	N/A	N/A	N/A	No	No	Yes	✓	✓	✓	✓
Rosenthal et al. [39]	MINLP	SS	No	IH	D	N/A	N/A	N/A	N/A	Yes	No	Yes	✓	✓	✓	✓
Tenpelmeyer [47]	MIP	SS	No	IH	S	N/A	N/A	N/A	N/A	No	No	Yes	✓	✓	✓	✓
Zhang & Zhang [51]	MIP	MS	No	IH	S	N/A	N/A	N/A	N/A	Yes	No	Yes	✓	✓	✓	✓
Haji et al. [22]	DP	MS	No	IH	D	S	S	S	S	Yes	No	Yes	✓	✓	✓	✓
Parlar & Perry [33]	MIP	MS	No	IH	D	N/A	S	S	S	No	No	Yes	✓	✓	✓	✓
Federguen & Yang [19]	NV	MS	No	IH	S	N/A	N/A	N/A	N/A	No	No	Yes	✓	✓	✓	✓
Ramasesh et al. [37]	OS	MS	No	IH	D	S	N/A	N/A	N/A	No	No	Yes	✓	✓	✓	✓
Rezaei & Davoodi [38]	MINLP	MS	No	MP	D	N/A	N/A	N/A	N/A	Yes	Yes	Yes	✓	✓	✓	✓
Mendoza & Ventura [29]	MIP	SS	No	IH	D	N/A	N/A	N/A	N/A	Yes	Yes	Yes	✓	✓	✓	✓
Burke et al. [11]	MIP	MS	No	IH	S	N/A	S	S	S	No	Yes	Yes	✓	✓	✓	✓
Yu et al. [50]	MIP	MS	No	IH	D	N/A	S	S	S	No	No	Yes	✓	✓	✓	✓
Dai & Qi [15]	MIP	MS	No	IH	D	N/A	S	S	S	No	No	Yes	✓	✓	✓	✓
Axsater et al. [5]	MIP	SS	Yes	IH	S	N/A	N/A	N/A	N/A	Yes	No	Yes	✓	✓	✓	✓
Sawik [42]	MIP	MS	No	MP	D	N/A	S	S	S	Yes	No	No	✓	✓	✓	✓
Hammami et al. [23]	MINLP	SS	No	MP	D	S	N/A	N/A	N/A	Yes	No	No	✓	✓	✓	✓
Ware et al. [49]	MINLP	SS	No	MP	D	N/A	N/A	N/A	N/A	Yes	Yes	Yes	✓	✓	✓	✓
Ventura et al. [48]	MINLP	SS	No	MP	D	N/A	N/A	N/A	N/A	Yes	Yes	Yes	✓	✓	✓	✓
Ruiz-Torres et al. [41]	MIP	MS	No	IH	D	N/A	S	S	S	Yes	No	No	✓	✓	✓	✓
Choudhary & Shankar [14]	MIP	SS	No	MP	D	N/A	N/A	N/A	N/A	Yes	Yes	Yes	✓	✓	✓	✓
Ekici [18]	MINLP	MS	No	IH	D	N/A	N/A	N/A	N/A	Yes	Yes	Yes	✓	✓	✓	✓
Mendoza & Ventura [30]	MINLP	MS	No	IH	D	N/A	S	S	S	Yes	Yes	Yes	✓	✓	✓	✓
Our Paper	MINLP	MS	Yes	IH	S	S	S	S	S	Yes	Yes	Yes	✓	✓	✓	✓

Table 1: A summary of supplier selection literature. (LT: Lateral Transshipment; DF: Disruption Frequency; DD: Disruption Downtime; CC: Contract Cost; PC: Procurement Cost; SC: Setup Cost; HC: Holding Cost; BC: Backorder Cost; Tr - C : Transportation Cost; S: Stochastic; D: Deterministic; OS: Order Statistics; LDT: Large Derivative Technique; GBD: Generalized Benders Decomposition; SS: Single Sourcing; MS: Multi Sourcing; IH: Infinite Horizon; SP: Single Period; and MP: Multi-Period.)

3. Model Formulation

3.1. Preliminaries

In this paper, we consider a two-stage supply chain network where the first stage represents $M \geq 2$ pre-qualified, candidate suppliers and the second stage represents $N \geq 2$ warehouses belonging to the same firm. Each selected supplier costs the firm a contractual fee, f_j , and a unit purchasing cost, c_j .

Additionally, each supplier $j \in \mathcal{J}$ offers the firm an annual capacity, W_j , and a random quality level, $0.5 \leq q_j \leq 1.0$. Specifically, the random quality in our setting is similar to random yield. In other words, supplier $j \in \mathcal{J}$ only delivers $q_j Q_{ij}$ when warehouse $i \in \mathcal{I}$ orders Q_{ij} . This percentage changes for each order placed with supplier j according to a random distribution. Supplier $j \in \mathcal{J}$ is susceptible to disruptions at frequency rate of θ_j and downtime rate of ν_j . The annual capacity of supplier j , W_j , does not take into account the disruptions.

Each warehouse faces an independent stationary, stochastic demand. For the sake of the computational analysis, we consider the demand is drawn from a Poisson distribution with mean λ_i , $i \in \mathcal{I} = \{1, \dots, N\}$.

To satisfy the demand at each warehouse i , $i \in \mathcal{I} = \{1, \dots, N\}$, the firm selects a subset of suppliers. The binary decision variable for supplier selection, X_j , is 1 if supplier j is selected, and it is 0, otherwise.

Next set of decision variables relate to the supplier-warehouse assignments and inventory replenishment. For warehouse i , $i \in \mathcal{I}$, and supplier j , $j \in \mathcal{J}$, variable Y_{ij} represents the total annual amount supplied by supplier j for warehouse i . We assume that each warehouse $i \in \mathcal{I}$ follows a continuous review reorder point-reorder quantity (nQ, R) policy for each associated source. Specifically, R_{ij} sets the reorder level for the inventory at warehouse i specifically sourced by supplier j , Q_{ij} represents the replenishment order by warehouse i from supplier j and n_{ij} is the smallest integer value that can bring the inventory position of the associated pair to their reorder level R_{ij} . Note that for each source, there is a separate R , which is referred as compartmentalization of inventory in the rest of the analysis. It is possible to manage the inventories without this compartmentalization, however, it creates an inferior solution as shown later. Note that $\frac{Y_{ij}}{n_{ij}Q_{ij}}$ is the expected number of orders per year between warehouse i and supplier j . To simplify the mathematical notation, we also define indicator decision variables A_{ij} and A_{ik}^T . That is, $\forall i \in \mathcal{I}$ and $\forall j \in \mathcal{J}$, $A_{ij} = \delta(Y_{ij}) = 1$, if warehouse i is sourced by supplier j ; it is set to 0, otherwise. Similarly, $\forall i \in \mathcal{I}$ and $\forall k \neq i \in \mathcal{I}$,

$A_{ik}^T = \delta(Y_{ik}^T) = 1$, if warehouse i is sourced by warehouse k , and it is set to 0, otherwise, where $\delta(\cdot)$ is an indicator function.

Warehouse i may also receive shipments from another warehouse k , $i \neq k$, $i, k \in \mathcal{I}$, where warehouse k essentially serves as an intermediate warehouse for the firm. The decision variables Y_{ik}^T , Q_{ik}^T , n_{ik}^T , and R_{ik}^T represent the annual sourced amount for warehouse i from warehouse k , each order quantity for warehouse i from warehouse k , integer value associated with the order quantity of the pair, and the reorder level for warehouse i 's inventory transshipped by warehouse k , respectively. Definition of both Y_{ij} and Y_{ik}^T ensures that each warehouse may receive multiple shipments from either multiple suppliers (i.e., *multi-sourcing*) and/or other warehouses (i.e., *lateral transshipments*), respectively. Each replenishment from a supplier incurs a fixed cost of K_i (or K_i^T , for orders from other warehouses) and an inventory carrying cost of h_i per unit per unit of time. Additionally, if a backlog occurs at warehouse i , a per unit per unit of time cost of s_i is incurred.

Once a warehouse places a replenishment order, the order arrives after a supplier-warehouse specific lead time, L_{ij} . We let L_{ik}^T represent the lead time for replenishment order of warehouse i from warehouse k . Due to disruptions, the actual observed lead time and corresponding lead time demand would be higher than the stated lead time. Furthermore, recalling the quality percentage q_j , the total amount of replenishment is $q_j n_{ij} Q_{ij}$ at the warehouse. This yield issue is the main reason for preferring an (nQ, R) policy over a typical (Q, R) policy due to the instability of the latter.

Our model explicitly captures the overall transportation costs corresponding to each order delivery between supplier $j \in \mathcal{J}$ and warehouse $i \in \mathcal{I}$ with two specific cost parameters: (i) a fixed cost, denoted by p_{ij} , which represents the sum of fixed costs of ordering (setup) and transportation; and (ii) a per-mile transportation cost, denoted by r_{ij} , which leads to a distance-based transportation cost of $r_{ij} d_{ij}$ per order delivery, where d_{ij} represents the distance between supplier $j \in \mathcal{J}$ and warehouse $i \in \mathcal{I}$ in miles. In case of transshipment order of warehouse i from warehouse k , transportation cost parameters include p_{ik}^T , r_{ik}^T and d_{ik}^T as the fixed cost, variable cost, and distance of the warehouses i and k , respectively. Problem setting is given in Figure 1.

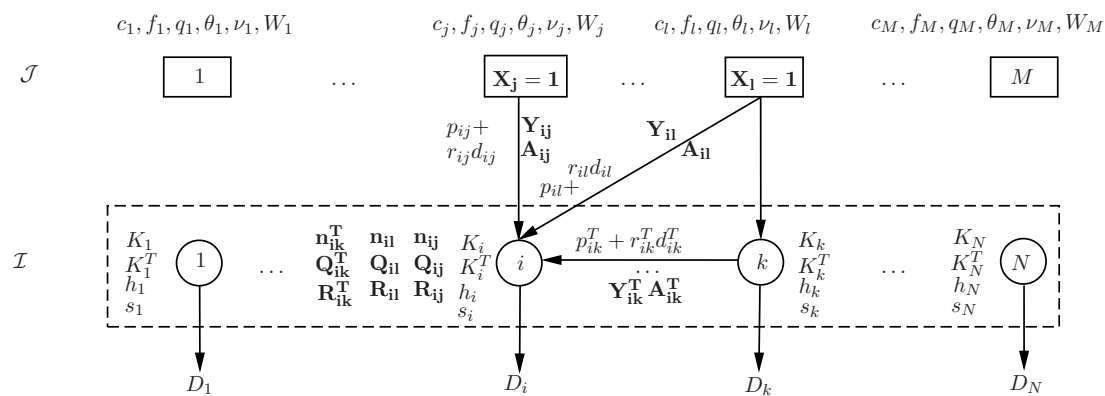


Figure 1: Problem setting.

1 A summary of the notation defined so far and additional notation is given in Table 2.

Parameters	Definitions
$\mathcal{I}$	warehouses of the firm, $\mathcal{I} = \{1, \dots, i, \dots, N\}$.
$\mathcal{J}$	set of candidate suppliers, $\mathcal{J} = \{1, \dots, j, \dots, M\}$.
c_j	unit procurement cost by supplier j , $\forall j \in \mathcal{J}$.
f_j	fixed annual contractual fee for supplier j , $\forall j \in \mathcal{J}$.
q_j	percentage of acceptable quality products by supplier j , $\forall j \in \mathcal{J}$.
θ_j	disruption frequency rate for supplier j , $\forall j \in \mathcal{J}$.
ν_j	disruption downtime rate for supplier j , $\forall j \in \mathcal{J}$.
W_j	annual throughput capacity of supplier j , $\forall j \in \mathcal{J}$.
p_{ij}	fixed transportation cost per replenishment to warehouse i , $\forall i \in \mathcal{I}$, from supplier j , $\forall j \in \mathcal{J}$.
r_{ij}	variable mileage cost per replenishment to warehouse i , $\forall i \in \mathcal{I}$, from supplier j , $\forall j \in \mathcal{J}$.
p_{ik}^T	fixed transportation cost per replenishment to warehouse i , $\forall i \in \mathcal{I}$, from warehouse k , $\forall k \in \mathcal{I}$.
r_{ik}^T	variable mileage cost per replenishment to warehouse i , $\forall i \in \mathcal{I}$, from warehouse k , $\forall k \in \mathcal{I}$.
d_{ij} (d_{ik}^T)	distance between warehouse i , $\forall i \in \mathcal{I}$, and supplier j (warehouse k), $\forall j \in \mathcal{J}$, $\forall k \in \mathcal{I}$.
L_{ij} (L_{ik}^T)	lead time between warehouse i , $\forall i \in \mathcal{I}$, and supplier j (warehouse k), $\forall j \in \mathcal{J}$.
$E[LT D_{ij}]$ ($E[LT D_{ik}^T]$)	expected lead time demand between warehouse i , $\forall i \in \mathcal{I}$, and supplier j (warehouse k), $\forall j \in \mathcal{J}$.
λ_i	demand arrival rate for warehouse $i \in \mathcal{I}$.
$F_i(\cdot)$	CDF of demand distribution for warehouse $i \in \mathcal{I}$.
K_i^T	fixed setup cost of warehouse i when sourced by other warehouses, $\forall i \in \mathcal{I}$.
K_i	fixed setup cost of warehouse i when sourced by suppliers, $\forall i \in \mathcal{I}$.
h_i	inventory holding cost per unit per time of inventory held at warehouse i , $\forall i \in \mathcal{I}$.
s_i	backorder cost per unit per time of stockout at warehouse i , $\forall i \in \mathcal{I}$.
Decision Variables	Definitions
X_j	1, if supplier j is selected, 0, otherwise; $\forall j \in \mathcal{J}$.
Y_{ij}	amount of warehouse i 's annual demand satisfied by supplier j , $\forall i \in \mathcal{I}$ and $\forall j \in \mathcal{J}$.
$A_{ij} = \delta(Y_{ij})$	1, if warehouse i is sourced by supplier j , 0, otherwise; $\forall i \in \mathcal{I}$ and $\forall j \in \mathcal{J}$.
Q_{ij}	order quantity of warehouse i from supplier j , $\forall i \in \mathcal{I}$, $\forall j \in \mathcal{J}$.
n_{ij}	integer value associated with Q_{ij} for warehouse i from supplier j , $\forall i \in \mathcal{I}$, $\forall j \in \mathcal{J}$.
R_{ij}	reorder point of warehouse i from supplier j , $\forall i \in \mathcal{I}$, $\forall j \in \mathcal{J}$.
Y_{ik}^T	amount of warehouse i 's demand satisfied by warehouse k , $\forall i, k \in \mathcal{I}$ and $i \neq k$.
$A_{ik}^T = \delta(Y_{ik}^T)$	1, if warehouse i is sourced by warehouse k , 0, otherwise; $\forall i \in \mathcal{I}$ and $\forall k \neq i \in \mathcal{I}$.
Q_{ik}^T	order quantity of warehouse i from warehouse k , $\forall i, k \in \mathcal{I}$, $i \neq k$.
n_{ik}^T	integer value associated with Q_{ik}^T for warehouse i from warehouse k , $\forall i, k \in \mathcal{I}$, $i \neq k$.
R_{ik}^T	reorder point of warehouse i from warehouse k , $\forall i, k \in \mathcal{I}$, $i \neq k$.

Table 2: Notation.

3.2. Model Development

The objective function of the integrated supplier selection and inventory problem includes the average annual costs associated with: supplier selection specific costs (contractual fees and procurement costs), warehouse inventory management costs (inventory ordering, holding, and shortage), and transportation related costs (loading/unloading and mileage) between selected suppliers and existing warehouses. Specifically, the objective function is given as

$$\sum_{j \in \mathcal{J}} f_j X_j + \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} c_j Y_{ij} \quad (1)$$

$$+ \sum_{i \in \mathcal{I}} \left[\sum_{j \in \mathcal{J}, Q_{ij} \neq 0} \frac{K_i Y_{ij}}{n_{ij} Q_{ij}} + \sum_{k \in \mathcal{I}, Q_{ik}^T \neq 0} \frac{K_i^T Y_{ik}^T}{n_{ik}^T Q_{ik}^T} \right] \quad (2)$$

$$+ \sum_{i \in \mathcal{I}} h_i \left[\sum_{j \in \mathcal{J}} A_{ij} \left\{ \frac{n_{ij} Q_{ij}}{2} + R_{ij} - E[LTD_{ij}] \right\} + \sum_{k \in \mathcal{I}, k \neq i} A_{ik}^T \left\{ \frac{n_{ik}^T Q_{ik}^T}{2} + R_{ik}^T - E[LTD_{ik}] \right\} \right] \quad (3)$$

$$+ \sum_{i \in \mathcal{I}} s_i \left[\sum_{j \in \mathcal{J}, Q_{ij} \neq 0} \eta(LTD_{ij}, R_{ij}) \frac{Y_{ij}}{n_{ij} Q_{ij}} + \sum_{k \in \mathcal{I}, Q_{ik}^T \neq 0} \eta(LTD_{ik}, R_{ik}^T) \frac{Y_{ik}^T}{n_{ik}^T Q_{ik}^T} \right] \quad (4)$$

$$+ \sum_{i \in \mathcal{I}} \left[\sum_{j \in \mathcal{J}, Q_{ij} \neq 0} \frac{(p_{ij} + r_{ij} d_{ij}) Y_{ij}}{n_{ij} Q_{ij}} + \sum_{k \in \mathcal{I}, Q_{ik}^T \neq 0} \frac{(p_{ik}^T + r_{ik}^T d_{ik}^T) Y_{ik}^T}{n_{ik}^T Q_{ik}^T} \right]. \quad (5)$$

In this objective function, (1) represents the contractual costs specific to the selected suppliers and average annual procurement costs. The inventory management costs are the average annual replenishment setup costs (2), average annual inventory holding costs (3), and average annual backorder costs (4). In (4), $\eta(\cdot, \cdot)$ represents the standardized loss function. Finally, the transportation costs for the warehouses from the suppliers and other warehouses in the network are captured by (5). As explained previously, for each replenishment order between warehouse $i \in \mathcal{I}$ and supplier $j \in \mathcal{J}$ (or warehouse $k \in \mathcal{I}$ and warehouse $i \in \mathcal{I}$), we incur p_{ij} (p_{ik}^T) for fixed dispatch, loading/unloading, and administrative costs and $r_{ij} d_{ij}$ ($r_{ik}^T d_{ik}^T$) for variable mileage costs. There will be, on average, $\frac{Y_{ij}}{n_{ij} Q_{ij}}$ ($\frac{Y_{ik}^T}{n_{ik}^T Q_{ik}^T}$) trips between supplier j (warehouse k) and warehouse i .

3.3. Related Constraints

The main constraints of this problem are:

- Demand at each warehouse should be satisfied via shipments from suppliers and/or other

warehouses:

$$\sum_{j \in \mathcal{J}} Y_{ij} + \sum_{k \in \mathcal{I}, k \neq i} Y_{ik}^T - \sum_{l \in \mathcal{I}, l \neq i} Y_{li}^T = E[D_i], \quad \forall i \in \mathcal{I}. \quad (6)$$

Constraint 6 states the material flow balance for each warehouse. The first two parts of the left hand side in this constraint are related to the material flow entering warehouse i and the last part is related to the departing flow from warehouse i .

- Warehouses may only be replenished from selected suppliers:

$$Y_{ij} \leq E[D_i]X_j, \quad \forall i \in \mathcal{I} \text{ and } \forall j \in \mathcal{J}. \quad (7)$$

- A supplier cannot satisfy the demand beyond its annual production capacity (without considering disruptions):

$$\sum_{i \in \mathcal{I}} Y_{ij} \leq W_j X_j, \quad \forall j \in \mathcal{J}. \quad (8)$$

- A supplier can be assigned to a warehouse only if it is selected:

$$A_{ij} \leq X_j, \quad \forall i \in \mathcal{I} \text{ and } \forall j \in \mathcal{J}. \quad (9)$$

- Relationships defined for the indicator variables $\mathbf{A}$ and $\mathbf{A}^T$:

$$\begin{aligned} Y_{ij} &\leq E[D_i]A_{ij}, & \forall i \in \mathcal{I} \text{ and } \forall j \in \mathcal{J}. \\ Y_{ik}^T &\leq E[D_i]A_{ik}^T, & \forall i, k \in \mathcal{I}, i \neq k. \end{aligned} \quad (10)$$

The overall model, even without the disruption consideration, is a stochastic MINLP. In fact, due to dependency issues and order crossings associated with multiple outstanding orders, it is not possible to obtain a closed form for the loss function $\eta(.,.)$. As others in the literature [15, 25, 27, 28], we use optimization via simulation to solve this problem.

Due to constraints (9) and (10), constraint set (7) is redundant; nevertheless, we keep them in the formulation as they are shown to be useful in the decomposition of the formulation in Section 4.1.

4. Solution Approach

Simulation-based optimization is a solution technique that has received attention for its ability to solve difficult problems, especially when the closed form solution of the objective function cannot

1 be obtained or evaluated. In our solution procedure, as shown in Figure 2, decomposition based
 2 heuristic algorithm searches the solution space to form trial solutions. Simulation model acts as a
 3 stochastic evaluator of the solution set passed to it. The best solution out of the set is reported
 4 back to the heuristic in order to refine the search space in the next iteration.

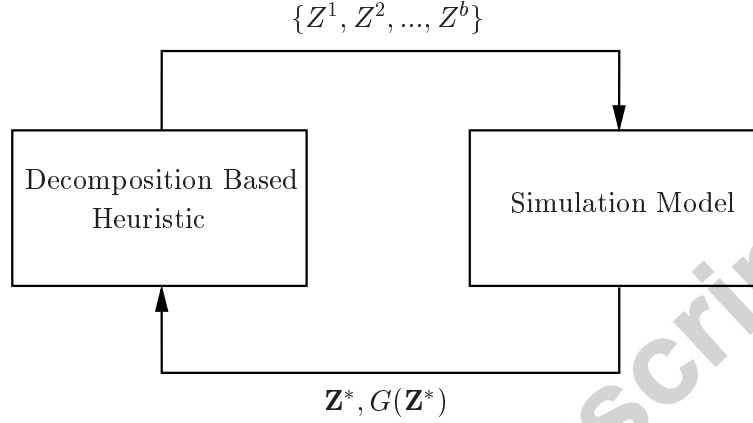


Figure 2: Overall solution methodology.

5 More specifically, our solution methodology is initiated by the decomposition based heuristic
 6 establishing an initial set of feasible solutions, each composed of vectors of decision variables per-
 7 taining to the supplier selection, demand assignment, and inventory management. The set of trial
 8 solutions, with cardinality of b , is then passed to the simulation model where a stochastic analy-
 9 sis with respect to the uncertainties in demand, disruption, and supplier quality is performed on
 10 each trial solution. The solution with the best performance measure (Z^*), i.e., lowest total cost
 11 ($G(Z^*)$), is returned to the search engine. The search engine then restarts the search with the
 12 best solution to generate the next set of trial solutions. This iterative procedure is terminated
 13 using an iteration limit termination condition. At this point, a resulting set of input parameters
 14 is obtained that optimizes (or nearly optimizes) the supply chain network cost performance. In
 15 other words, a solution in the form of a set of selected suppliers ($\mathbf{X}$), warehouse to supplier and
 16 warehouse to warehouse demand assignments ($\mathbf{Y}, \mathbf{A}$ and $\mathbf{Y}^T, \mathbf{A}^T$), and warehouse inventory policy
 17 decisions ($\mathbf{n}, \mathbf{Q}, \mathbf{R}, \mathbf{n}^T, \mathbf{Q}^T, \mathbf{R}^T$) is obtained that minimizes the total network cost while adhering
 18 to operational constraints (i.e. capacity and demand).

4.1. Development of the Decomposition based Heuristic

Our decomposition heuristic starts with a supplier selection metaheuristic that determines the selected suppliers $\mathbf{X}$. However, a complete solution $\mathbf{Z}$ for the problem includes all of the decision variables $\{\mathbf{X}, \mathbf{Y}, \mathbf{n}, \mathbf{Q}, \mathbf{R}, \mathbf{Y}^T, \mathbf{n}^T, \mathbf{Q}^T, \mathbf{R}^T\}$. Similar to Keskin et al. [25] we obtain $\mathbf{X}$ through a hybrid scatter search implementation, however, as opposed to Keskin et al. [25], we generate the remaining decision variables via decomposition of the problem into generalized transportation and generalized inventory optimization as shown in Figure 3.

4.1.1. Generalized Transportation Problem: Determining $\mathbf{Y}$ and $\mathbf{Y}^T$

Given the selected suppliers, $\mathbf{X}$, we determine the assignment of warehouses demands to suppliers, $\mathbf{Y}$, and other warehouses, $\mathbf{Y}^T$, by solving the following subproblem:

$$\min_{\mathbf{Y}, \mathbf{Y}^T} \sum_{i \in \mathcal{I}} \left[\sum_{j \in \mathcal{J}} (K_i + p_{ij} + r_{ij} d_{ij}) \frac{Y_{ij}}{\bar{n}_{ij} \bar{Q}_{ij}} + \sum_{k \in \mathcal{I}} (K_i^T + p_{ik}^T + r_{ik}^T d_{ik}^T) \frac{Y_{ik}^T}{\bar{n}_{ik}^T \bar{Q}_{ik}^T} \right] \quad (TrSubProblem)$$

subject to

$$\begin{aligned} \sum_{j \in \mathcal{J}} Y_{ij} + \sum_{k \in \mathcal{I}, k \neq i} Y_{ik}^T - \sum_{l \in \mathcal{I}, l \neq i} Y_{li}^T &= E[D_i], & \forall i. \\ Y_{ij} &\leq E[D_i] \bar{X}_j, & \forall i \in \mathcal{I} \text{ and } \forall j \in \mathcal{J}. \\ \sum_{i \in \mathcal{I}} Y_{ij} &\leq W_j \bar{X}_j, & \forall j \in \mathcal{J}. \\ Y_{ij} \text{ and } Y_{ik}^T &\geq 0, & \forall i \text{ \& } k \in \mathcal{I} \text{ and } \forall j \in \mathcal{J}. \end{aligned}$$

Note that the *TrSubProblem* formulation needs the order quantities $\bar{\mathbf{n}}\bar{\mathbf{Q}}$ and $\bar{\mathbf{n}}^T\bar{\mathbf{Q}}^T$ of each warehouse. Initially, within each iteration of scatter search, while determining $\mathbf{Y}$ and $\mathbf{Y}^T$, we use the economic order quantity (EOQ) to estimate the equally dispersed order quantities of each warehouse from any other entity in the system, i.e., for each warehouse $i \in \mathcal{I}$,

$$\bar{Q}_{ij} = \frac{\sqrt{\frac{2K_i E[D_i]}{h_i}}}{M} \quad \text{and} \quad \bar{Q}_{ik}^T = \frac{\sqrt{\frac{2K_i^T E[D_i]}{h_i}}}{N-1},$$

where M and N are the number of suppliers and warehouses, respectively. We set $\bar{n}_{ij}$ and $\bar{n}_{ik}^T$ equal to 1, as in a typical (Q, R) policy. Note that initializing the order quantity for each supplier or warehouse with pure EOQ overestimates the real order quantity since it assumes the whole demand is satisfied by one location. By scaling the pure EOQ with the number of suppliers and the number

of warehouses, we get a better estimate of the number of trips between locations and hence, the transportation cost.

Also note that we iterate between generalized inventory optimization and transportation problem so that we reach a convergence for the values of $\mathbf{Y}$, $\mathbf{Y}^T$, $\mathbf{nQ}$, and $\mathbf{n}^T\mathbf{Q}^T$ among consecutive iterations. We observed that, typically, within five iterations, this convergence is achieved.

4.1.2. Generalized Inventory Optimization: Determining $\mathbf{n}$, $\mathbf{Q}$, $\mathbf{R}$, $\mathbf{n}^T$, $\mathbf{Q}^T$, and $\mathbf{R}^T$

Once the network decisions are determined, we need to determine the optimal $(\mathbf{nQ}, \mathbf{R})$ and $(\mathbf{n}^T\mathbf{Q}^T, \mathbf{R}^T)$. As opposed to Keskin et al. [25], we propose the following approximation since the exact solution for this inventory subproblem is computationally demanding, as elaborated in Hadley and Whitin [21], especially when this subproblem is run in every iteration of the generalized inventory optimization module within each replication of the supplier selection search. As shown in Figure 3, after obtaining $\bar{Y}$ and $\bar{Y}^T$ from *TrSubProblem*, by solving an *InvProblem* for each warehouse i , we calculate the corresponding order quantities as:

$$Q_{ij} = \sqrt{\frac{2\bar{Y}_{ij} [K_i + (p_{ij} + r_{ij}d_{ij})]}{h_i}}, \quad \forall i \in \mathcal{I} \ \& \ \forall j \in \mathcal{J}, \quad (11)$$

$$Q_{ik}^T = \sqrt{\frac{2\bar{Y}_{ik}^T [K_i^T + (p_{ik}^T + r_{ik}^Td_{ik}^T)]}{h_i}}, \quad \forall i \in \mathcal{I} \ \& \ \forall k \neq i \in \mathcal{I}, \quad (12)$$

We emphasize further that compartmentalization of the inventory decision variables is a requirement for the multi-sourcing environments. First, it is necessary to have different order quantities from different sources since the order quantity resolves the trade-off with regards to fixed ordering and inventory holding costs. When a large portion of the fixed ordering costs are due to transportation costs among the supplier and the warehouse, order quantities become specific to each supplier-warehouse (warehouse-warehouse) combination. Second, the reorder point is determined using the lead time demand and safety stock, both of which are influenced by the supply source. The lead time of a source, the order quantity from that source, and the portion of annual demand satisfied by that source contribute to the calculation of the reorder point. Ignoring all of these factors and offering a single reorder point is an over-simplification of the problem. Following Mendoza and Ventura [29], Mendoza and Ventura [30], and Pazhani et al. [35], we also offer the compartmentalized approach to most effectively capture the benefits of multi-sourcing.

In order to accurately calculate the reorder levels, we use a service level based calculation, as opposed to shortage cost based optimization in Keskin et al. [25]. In essence, warehouse i , $i \in \mathcal{I}$,

has one safety stock that is fulfilled by a number of suppliers and other warehouses that source this warehouse. For a specific in-stock service level (Type I) α , we calculate the reorder point of warehouse i as $R_i = F_i^{-1}(\alpha)$. Once R_i is determined, it is rationed according to the Y_{ij} and Y_{ik}^T values. More explicitly, we determine a weight for each supply source:

$$\begin{aligned}\Psi_{ij} &= \frac{Y_{ij}}{\sum_{j \in \mathcal{J}} Y_{ij} + \sum_{k \in \mathcal{I}} Y_{ik}^T} & \text{for } j \in \mathcal{J} \text{ and } Y_{ij} > 0; \\ \Psi_{ik}^T &= \frac{Y_{ik}^T}{\sum_{j \in \mathcal{J}} Y_{ij} + \sum_{k \in \mathcal{I}} Y_{ik}^T} & \text{for } k \in \mathcal{I} \text{ and } Y_{ik}^T > 0.\end{aligned}$$

Then, $R_{ij} = \Psi_{ij} R_i + \lambda_i L_{ij}$ and $R_{ik}^T = \Psi_{ik}^T R_i + \lambda_i L_{ik}^T$.

As stated in Zipkin [52], the (Q, R) policy is unstable when quality issues are involved with the ordered quantity $\mathbf{Q}$. The use of (nQ, R) policy, ensures the stability of such system in the long run. Definition of n_{ij} and n_{ik}^T ensures the stability conditions of the simulation model. Specifically, n_{ij} (n_{ik}^T) is the smallest non-negative integer value that once multiplied by Q_{ij} (Q_{ik}^T) can bring the inventory position of the pair to or above the reorder point R_{ij} (R_{ik}^T).

Note that this inventory subproblem, by itself, is quite complicated. Investigation of its solution with other inventory policies and via other solution approaches is a significant research direction.

4.1.3. Overall Optimization Algorithm

Figure 3 shows the iterative algorithm performed within the decomposition based heuristic. As explained earlier, *InvProblem* and *TrSubProblem* are run iteratively until a convergence condition is met for each set of selected suppliers.

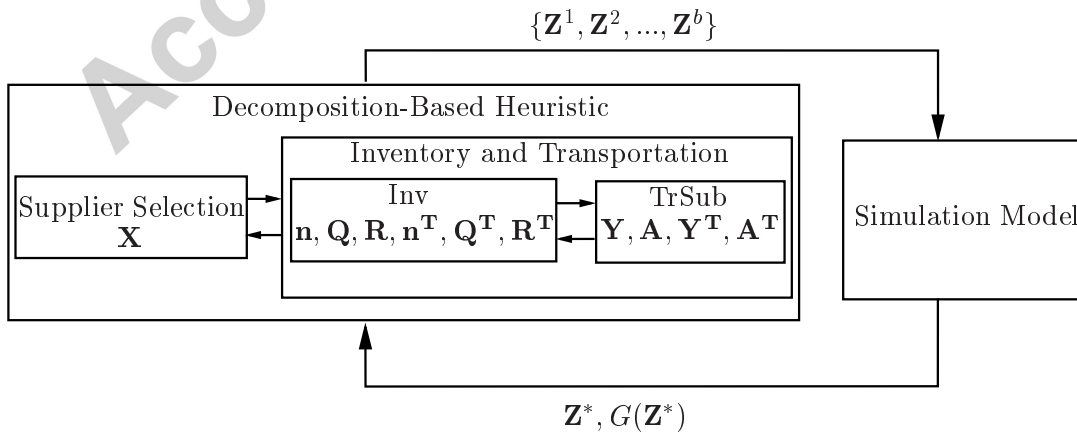


Figure 3: Detailed solution methodology.

Once $\mathbf{Z} = \{\mathbf{X}, \mathbf{Y}, \mathbf{n}, \mathbf{Q}, \mathbf{R}, \mathbf{Y}^T, \mathbf{n}^T, \mathbf{Q}^T, \mathbf{R}^T\}$ is generated, these decisions are sent to the simulation module for objective function evaluation.

4.2. Simulation Framework

Our discrete-event simulation has five main modules, each coded in C++. The modules are i) network structure and initialization; ii) order placement; iii) inventory adjustment; iv) main simulator, and v) cost calculation. Information exchange (creating events) between these five modules is shown, as a flowchart, in Figure 4.

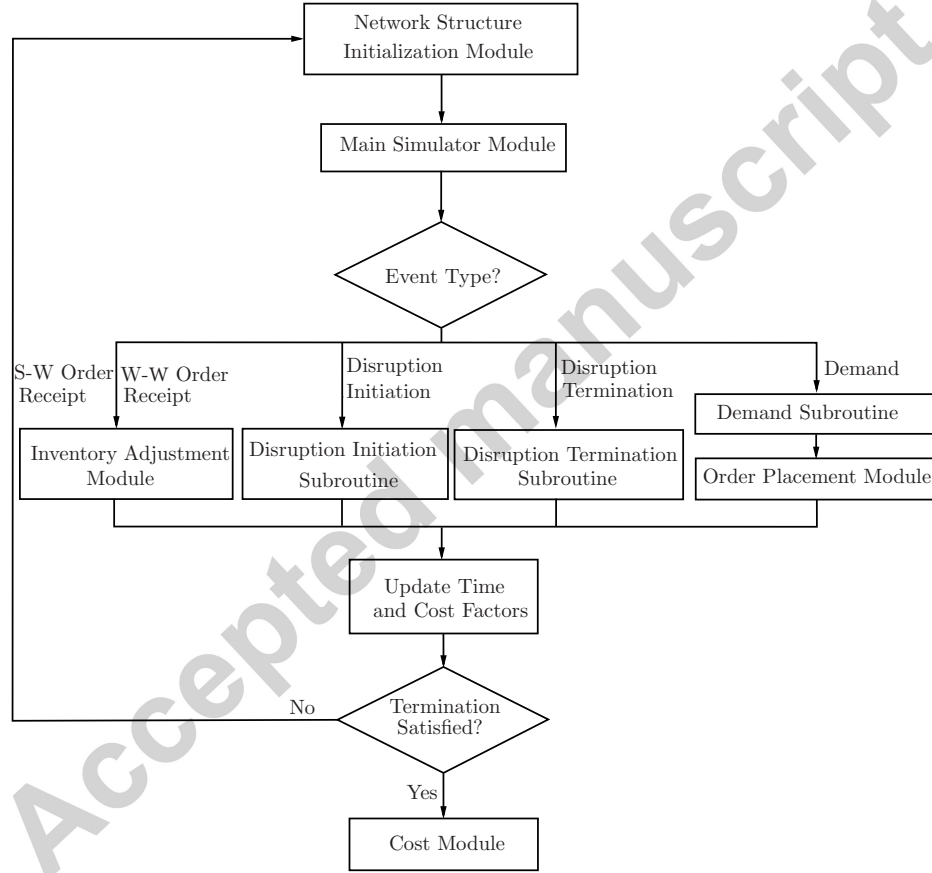


Figure 4: Simulation model.

In Figure 4, a single replication of our simulation model is depicted. Each time the simulation reads the next event via main simulator module and required *information* is obtained from this event. Based on the event type, and the information stored in the event, the corresponding module or sub-routine is called to process this event. There are five types of event in our simulation model. Demand, Supplier-Warehouse (S-W) order receipt, Warehouse-Warehouse (W-W) order receipt,

disruption initiation, and disruption termination are explicitly modelled in the simulation. Each one of these events initiate each other by means of sub-routines and modules. Since inventory holding and shortage costs are time-phased averages in our model, they are updated after each event using the time increment and corresponding cost parameters. The process of generating new events needs to be interrupted at a point by a termination condition. This termination condition is a maximum run time period and a maximum replication number in our simulation model, i.e., total simulation time and number of replications. When process is terminated, the cost module calculates the corresponding cost factors and reports them along with the total cost of simulation.

Module 1: Network Structure and Initialization

In this module, we set up the supply chain network by specifying the number of suppliers and warehouses as well as all other warehouse-related and supplier-related data. In addition, this module is responsible for setting up the decision variables based on the outcome of the decomposition based heuristic.

Module 2: Order Placement

An order from a warehouse could be placed with a supplier (S-W) or with another warehouse (W-W). The inventory levels fluctuate throughout the simulation due to one of the five events mentioned earlier. In order to capture the value of multi-sourcing, where multi-sourcing is allowed, we choose a pair (W-W or S-W) with the maximum inventory to satisfy the unit demand. It is noted that, as reflected in Section 4.1.2, the compartmentalization of the inventory is necessary in order to be able to capture the value of multi-sourcing. Even though there could be other demand satisfaction rules, choosing the source with the maximum inventory reduces both the inventory holding and backorder costs. Additionally, inventory position is routinely checked and compared to the compartmentalized reorder level of each pair to determine if an order should be placed for that pair whenever it is reduced through demand satisfaction or transshipments. Different order quantities dictate the usage of different inventory level and positions, which in turn also justifies the requirement for separate reorder levels for each pair of the network. If the shipper of an order is a supplier, an S-W order receipt event will be scheduled. On the other hand, if the shipper is another warehouse (transshipment option), a W-W order receipt event will be potentially scheduled. For lateral transshipment considerations, the inventory position of the W-W pair is also compared to its reorder level at demand events (and transshipments) to determine the necessity of order placement between the pair. However, placing an order from a warehouse by another warehouse is more

complicated than from a supplier. When an order is placed with a warehouse, i.e., a supplying warehouse, in addition to checking the outstanding orders list of that warehouse, it is necessary to ensure the *supplying warehouse* is capable of the transshipment. The transshipment will be approved only if it is cost-wise effective when compared to other options including multi-sourcing, single sourcing, inventory holding, and backordering. This check is run through the simulation model with consideration of the stochastic demand, disruptions, and supplier qualities.

Module 3: Inventory Adjustment

An order could arrive from a warehouse or from a supplier. When the order comes from a supplier, physical inventory of the corresponding receiving warehouse is updated. Otherwise, the inventory position of the shipping warehouse and physical inventories of both warehouses are updated.

Module 4: Main Simulator

This module is the heart of the discrete events simulation. It reads the future events list and extracts the information of next event to call the proper module or sub-routine. There are five different events: i) demand generation; ii) disruption generation; iii) disruption downtime generation; iv) S-W order receipt; and v) W-W order receipt.

When a unit demand event occurs for a warehouse, the demand sub-routine processes that event. As noted in module 2, the unit demand is satisfied from the source with the maximum inventory. Physical inventory and inventory position of this warehouse are decremented by the demand. Within this subroutine, we check whether it is necessary to place an order with the corresponding source (either a supplier or a warehouse), in which case module 2 processes this order placement.

Disruption frequencies and disruption down times are controlled by the disruption termination and disruption initiation sub-routines, respectively, for each supplier. When an event of disruption initiation type occurs for a supplier, a disruption downtime is generated and a disruption termination event is scheduled for that supplier. On the other hand, when a disruption termination event occurs for a disrupted supplier, a time for the next disruption event is generated and a disruption initiation event will be scheduled.

Event types S-W and W-W order receipts are processed by module 3 as explained. In cases with disruption consideration, when module 2 initiates an order for a S-W pair with respect to their inventory position and reorder level, an order receipt event may not be placed in the future events list immediately. This happens when the corresponding supplier either i) is disrupted, or ii) does not have sufficient inventory. Such orders enter into the outstanding orders list. When

each supplier comes out of disruption and has sufficient inventory to satisfy the first element of the list, an order receipt may then be scheduled. Disruptions pose as an additional delay for an order accordingly. If a particular source is disrupted, its inventory will likely not be the maximum one. Hence, this choice also reduces the possibility and the cost of backorders. Another possibility to avoid backorders is to invoke emergency lateral transshipments (i.e., reactive transshipments) in addition to proactive lateral transshipments. We leave this as a possible future research direction.

An important stage of any simulation is its validation. We validate our simulation using the mixed integer programming formulation without uncertainty and disruptions. For this purpose, we compare the cost components of the analytical model with the output of our simulation model.

5. Computational Experiments

In this section, we demonstrate the performance of the solution method in terms of solution quality and computational speed via extensive experiments. The decomposition based heuristic including supplier selection, inventory, and transportation subproblems are coded in C++. The transportation subproblem calls CPLEX 12.6 for optimizing the shipment decisions. The experiments are run on a machine with 2.66 GHz processor and 4GB of RAM.

5.1. Experimental Setup

We consider a firm with 15 warehouses while having a 10-supplier base. Simulation module evaluates each set of decision variables with 30 replications, each replication having a length of 5 years. There are eight hours per day and 250 days per year. We calculate the distance between warehouses and suppliers using the Euclidean (straight-line) distance among coordinates P_i and P_j , $i \in \mathcal{I}$ and $j \in \mathcal{J}$,

$$d_{ij} = d(P_i, P_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}.$$

Similarly, between warehouses, we use the straight-line distance among P_i and P_k , $i \in \mathcal{I}$ and $k \in \mathcal{J}$,

$$d_{ik}^T = d(P_i, P_k) = \sqrt{(x_i - x_k)^2 + (y_i - y_k)^2}.$$

Both distance measures, d_{ij} and d_{ik}^T , directly influence the lead time calculation. We use the uniform distribution to generate random data for the network parameters with given data ranges. Warehouse i experiences average unit demand arrivals with rate λ_i . Although our solution methodology can be applied to any type of demand distribution, Poisson demand is considered for computational

Table 3: Base case parameters.

	Parameters	Values	Units
Warehouses, $i \in \mathcal{I}$	Warehouse Location	$[U(0.00, 500.00), U(0.00, 500.00)]$	
	λ_i	$U(0.50, 1.50)$	units/hour
	h_i	$U(0.50, 3.00)$	\$/ (unit $\times$ year)
	s_i	$U(2.50, 5.00)$	\$/ (unit $\times$ year)
	K_i^T	$U(700.00, 1200.00)$	\$/order
	K_i	$U(500.00, 1000.00)$	\$/order
Suppliers, $j \in \mathcal{J}$	Supplier Location	$[U(0.00, 500.00), U(0.00, 500.00)]$	
	c_j	$U(0.40, 2.00)$	\$/unit
	f_j	$U(50, 000, 100, 000)$	\$/supplier
	q_j	$U(0.60, 1.00)$	
	θ_j	$U(1.00, 7.00)$	days
	ν_j	$U(0.50, 2.00)$	days
(Warehouse , Supplier), $i \in \mathcal{I}, j \in \mathcal{J}$	W_j	$U(7500.00, 10000.00)$	units/year
	L_{ij}	$[\frac{U(1.00, 2.00)}{60}] * d_{ij}$	hours
	r_{ij}	$U(0.75, 3.00)$	\$/mile
(Warehouse , Warehouse), $i \neq k \in \mathcal{I}$	p_{ij}	$U(250.00, 500.00)$	\$/order
	L_{ik}^T	$[\frac{U(1.00, 2.00)}{60}] * d_{ik}^T$	hours
	r_{ik}^T	$U(0.50, 1.50)$	\$/mile
	p_{ik}^T	$U(125.00, 250.00)$	\$/order

analysis. Using 8 hours as work hours per day and 250 days as work days per year, the expected annual demand and average lead time demand are estimated as $E[D_i] = \lambda_i \times 8 \times 250$ and $E[LT D_{ij}] = \lambda_i \times L_{ij}$ ($E[LT D_{ik}^T] = \lambda_i \times L_{ik}^T$), respectively. Other problem parameters used to generate the base case network are shown in Table 3. Note that these experimental settings are consistent with the ones in the literature (see [25],[26], [29], and [30]).

We perform two sets of experiments. First set is the disruption related experiments and the second set is parameter sensitivity related experiments. Each set of experiments is performed with three network structures: (1) both multi-sourcing and lateral transshipment are possible (MS-LT); (2) only single sourcing is possible (SS); and (3) multi-sourcing is allowed but transshipment between warehouses is not (MS-NLT). Note that SS and MS-NLT networks, when compared to the MS-LT network, represent the value of multi-sourcing and lateral transshipment in the supply chain network, respectively.

5.2. Supplier Disruption Experiments: Experimental Results

Disruption related experiments aim to investigate the effect of various disruption scenarios on each network structure performance. We performed six distinct disruption test cases specified on Table 4 with each of the three network structures specified in Section 5.1.

Table 4: Disruption experimental test cases.

Test Case	Explanation	Parameters Changed
Base Case	frequent and short disruptions	$\theta_j = U(1.00, 7.00), \nu_j = U(0.50, 2.00)$.
Test Case 1	rare and long disruptions	$\theta_j = U(40.00, 50.00), \nu_j = U(5.00, 10.00)$.
Test Case 2	$\frac{1}{2}$ Base Case & $\frac{1}{2}$ Test Case 1	$P(\theta_j = U(40.00, 50.00), \nu_j = U(5.00, 10.00)) = 0.50;$ $P(\theta_j = U(1.00, 7.00), \nu_j = U(0.50, 2.00)) = 0.50$.
Test Case 3	split freq & short based on unit cost	$c_j < 1.2 : \theta_j = U(1.00, 3.00), \nu_j = U(1.00, 2.00);$ $c_j \geq 1.2 : \theta_j = U(3.00, 7.00), \nu_j = U(0.50, 1.00)$.
Test Case 4	split rare & long based on unit cost	$c_j < 1.2 : \theta_j = U(30.00, 45.00), \nu_j = U(8.00, 10.00);$ $c_j \geq 1.2 : \theta_j = U(45.00, 60.00), \nu_j = U(5.00, 7.00)$.
Test Case 5	freq & short for low c_j rare and long for high c_j	$c_j < 1.2 : \theta_j = U(1.00, 7.00), \nu_j = U(0.50, 2.00);$ $c_j > 1.2 : \theta_j = U(40.00, 50.00), \nu_j = U(5.00, 10.00)$.

In Figure 5, using a radar chart, we present the results for all of these six disruption test cases. The total cost of each network structure is shown with a distinct line style and being closer to the center of the radar implies a lower total cost. We compare the total cost of MS-LT with SS and MS-NLT networks. The numbers written in the parentheses in Figure 5 are percentage of savings obtained from multi-sourcing and lateral transshipments (MS-LT) compared to single sourcing (SS) and not allowing lateral transshipments (MS-NLT), respectively. In particular, we calculate the percentage difference between MS-LT and SS (or MS-NLT) as follows:

$$\% \text{improvement with MS-LT over SS (MS-NLT)} = \frac{C_{SS(MS-NLT)} - C_{MS-LT}}{C_{MS-LT}} \times 100,$$

where $C_{(*)}$ represents the cost of the corresponding $(*)$ network.

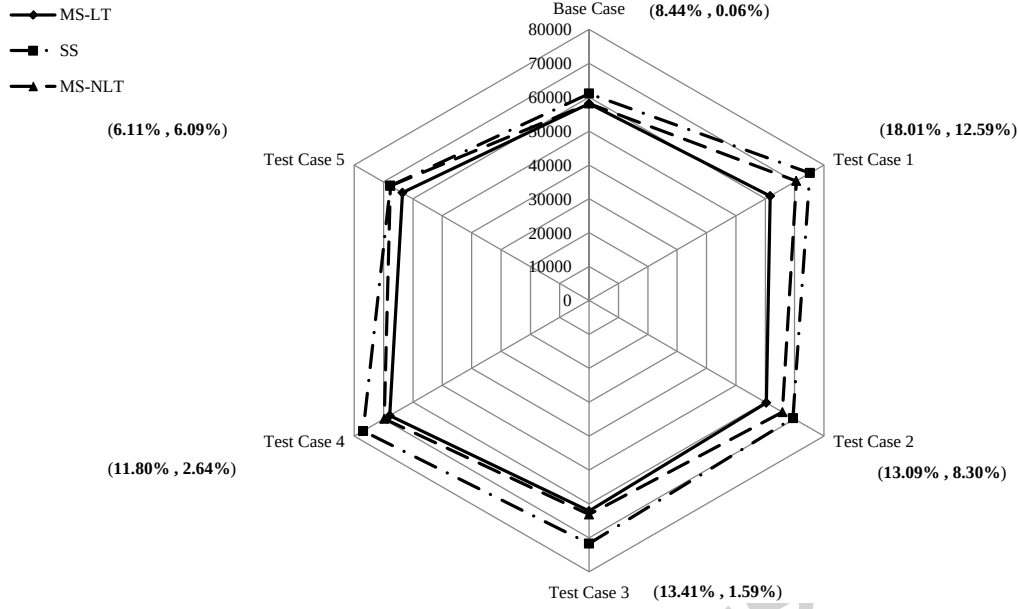


Figure 5: Total costs in supplier disruption experiments.

Figure 5 confirms cost ordering as MS-LT, MS-NLT, and then SS, where MS-LT has the lowest cost, and SS the highest. Essentially, as observed in theory and practice, as the flexibility in the network structure increases, the cost benefits follow. Additionally, we note that in all of the test cases, frequent and short disruptions result in a better overall total cost of the system regardless of the network type. This result is also in agreement with the general belief that rare but long disruptions have an adverse impact on the total system cost.

However, more interesting results are related to the relative difference between these three networks under these six test cases. In base case, there is almost no difference between the MS-LT and MS-NLT. Hence, with frequent but short disruptions, considering the difficulty of implementing proactive lateral transshipments, it is not desirable to include them just for cost benefits. In test case 1, all of the network design alternatives have higher costs. However, rare and long disruptions impact the SS network the most. Therefore, the overall savings for using MS-LT is the largest for this case. A similar analysis holds for the savings obtained from the lateral transshipments.

Based on the settings in test cases 3 and 4, the supplier base is “mixed” where each supplier has one “good” quality and one “bad” quality. For instance, in test case 3, a supplier with lower unit cost

has rare and longer disruptions whereas a supplier with higher unit cost has frequent disruptions with shorter durations. Due to this diversity, the cost benefits from lateral transshipments are diminished. It is always possible to find several favorable suppliers with desired properties. Hence, the difference between MS-LT and MS-NLT is diminished in these cases. However, SS cannot take advantage of this diversity since it is limited to selecting only one supplier.

Conversely, in test case 5, the disruption characteristics are also based on unit cost, however, the supplier base is not as diverse. In test case 5, SS and MS-NLT perform similarly in terms of cost where SS has a better performance compared to other cases and MS-NLT has a worse one. This finding is interesting as when the unit cost is split according to the disruption characteristics, SS can offer a competitive solution as compared to MS-NLT and MS-LT. In fact, the cost difference between MS-LT and the other two networks appears due to existence of lateral transshipments. All of these cases highlight the situations where it would be useful to consider more flexibility in the network structure given different disruption characteristics.

We investigate the contribution of operational cost factors in different network structures. We define the operational cost composed of all the previous cost components defined in Section 3.2 excluding the contractual cost in Equation 1 due to its comparably bigger magnitude than other cost factors. We consider two specific test cases: base case and test case 1 (frequent and short disruptions versus rare and long disruptions). We compare the operational cost division of these test cases for each network structure in Figure 6.

As a general observation, as we move from base case to test case 1, the contribution of backorder costs decreases while that of the procurement costs increases. Even though one may expect that with rare and long disruptions, the backorder costs would be higher. However, in this case, we find the firm more prepared to hedge against this type of disruption. Long disruption downtimes force the firm to buy in larger quantities to hedge against the uncertainty arising from the disruptions and in turn, decrease the contribution of backorder cost in the total operational cost. With respect to the changes in disruption characteristics, the other cost components do not vary as much.

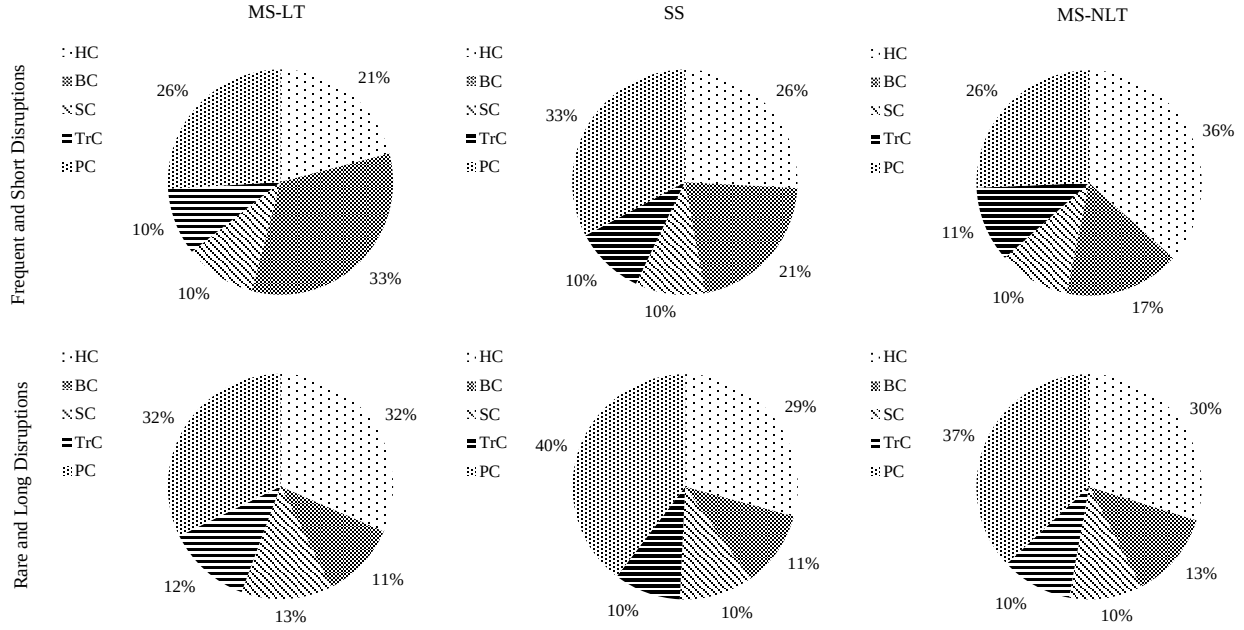


Figure 6: Operational cost contributors in selected supplier disruption experiments.

Within the base case, with short and frequent disruptions, as we move from the MS-LT network towards SS and MS-NLT networks, the contributions of holding cost and backorder cost decreases and that of purchasing cost increases. In both SS and MS-NLT networks, to satisfy the same demand structure as in an MS-LT network, the supply chain ends up buying more units of product from more reliable but expensive suppliers as the possibility of back up supplying has been omitted or diminished significantly. We observe a similar trend for test case 1 with rare and long disruptions.

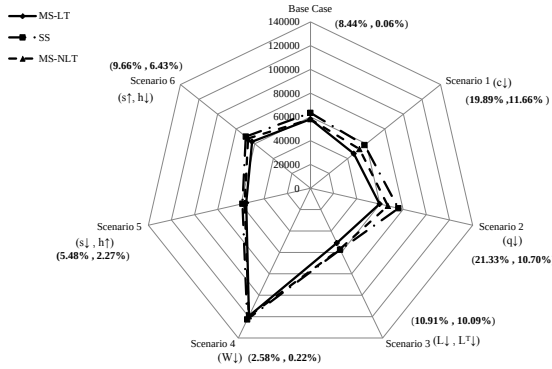
5.3. Parameter Sensitivity Experiments

In this section, we test the sensitivity of the results with regards to a number of problem parameters. Specifically, we experiment on a total number of six scenarios across two disruption settings suggested by base case and test case 1 in Section 5.2, representing frequent and short versus rare and long disruptions. Our goal is to investigate the sensitivity of each disruption setting towards the changes in the network parameters in this section. The details of each scenario are further explained in Table 5. We note that we choose these six scenarios as unit costs, supplier quality and capability, lead times, and inventory parameters are quoted as the most significant contributors of supplier selection.

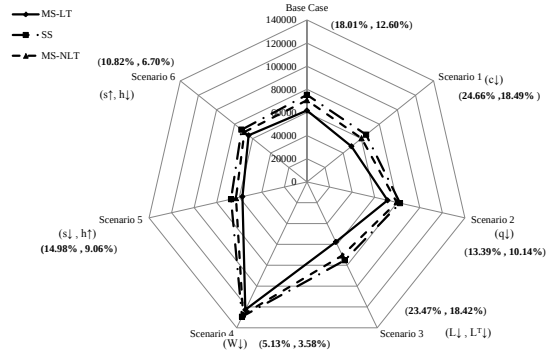
Table 5: Parameter sensitivity experimental scenarios

Scenario	Explanation	Parameters Changed
Scenario 1	lower unit cost	$c_j : U(0.04, 0.20)$
Scenario 2	lower supplier qualities	$q_j : U(0.50, 0.75)$
Scenario 3	lower lead times	$L_{ij} : [\frac{U(0.50, 1.00)}{60}] * d_{ij}$ $L_{ik}^T : [\frac{U(0.50, 1.00)}{60}] * d_{ik}^T$
Scenario 4	lower supplier capacities	$W_j : U(15.00, 20.00)$
Scenario 5	lower backorder & higher holding cost	$s_i : U(1.25, 2.50) \& h_i : U(1.00, 6.00)$
Scenario 6	higher backorder & lower holding cost	$s_i : U(5.00, 10.00) \& h_i : U(0.25, 1.50)$

Figure 7 depicts a comparison among these scenarios in terms of total cost.



(a) Frequent and short disruptions



(b) Rare and long disruptions

Figure 7: Total cost in parameter sensitivity experiments

Each scenario, exploring the sensitivity of the results towards a specific parameter within the parenthesis, is tested with three different network structures in both Figures 7(a) and 7(b). Percentage savings with MS-LT are again given inside the parenthesis compared to SS and MS-NLT, respectively.

In both Figures 7(a) and 7(b), as an overall expected result, the MS-LT network still performs considerably better than SS and MS-NLT networks. We observed that decreasing the capacity of the suppliers has the most detrimental effect on the supply chain network performance regardless of the network structure and disruption setting. Since the contractual cost of the suppliers is one of the main contributors to the total cost (see Section 5.2), any changes to the capability of the

suppliers has a considerable negative effect on the total cost of the supply chain network. This is due to the infeasibility of the solutions with same number of suppliers as in the base case, since the capacity of the suppliers is not sufficient for satisfying the overall demand of the warehouses. Therefore, the algorithm resorts to solutions with more suppliers (or more expensive suppliers with higher capacity), and ultimately, increasing the overall cost. This observation is also evidenced in the literature in [45].

Within both Figures 7(a) and 7(b), in scenario 1, decreasing unit cost improves the total cost in all networks, however, the largest improvement comes from the MS-LT network. Decreasing the quality offered by the suppliers, in scenario 2, increases the total cost of the solutions in all of the networks, but the increase is minimal with MS-LT network compared to the other two networks. The same pattern repeats when lead times are decreased in scenario 3. This result implies that changes in unit cost, quality and lead time in supply chains can be handled by an MS-LT network better than an SS or MS-NLT network. MS-LT network is more robust and can take advantage of the problem settings in finding the best solution (see Section 5.3.1).

In both Figures 7(a) and 7(b), when two parameters with reverse effects change simultaneously, (e.g., scenarios 5 and 6), the analysis becomes more complicated. Although in both scenarios 5 and 6, the MS-LT network results in the best overall cost, but the MS-NLT network in scenario 5 results in higher decrease in total cost compared to its total cost under the base case. In scenario 6, all network settings result in higher costs. This implies that, mostly in supply chain networks with only single sourcing possibility, higher cost of backordering dominates and adversely affects the total cost even though the cost of inventory holding is decreased. This finding is consistent with findings in the literature.

Comparing each scenario in Figure 7(a) with the corresponding ones in Figure 7(b), we realize that the total cost increases in all of the scenarios. This result was already stated as the preference of the frequent and short disruptions to rare and long disruptions in Section 5.2. However, the more interesting outcome relates to the comparison of the savings from MS-LT when disruptions are rare and long in Figure 7(b) to the savings in Figure 7(a) when disruptions are short and frequent. Specifically, in supply chains where disruptions are rare and long, savings from both multi-sourcing and lateral transshipment are considerable, even though all of the network configurations have higher overall costs.

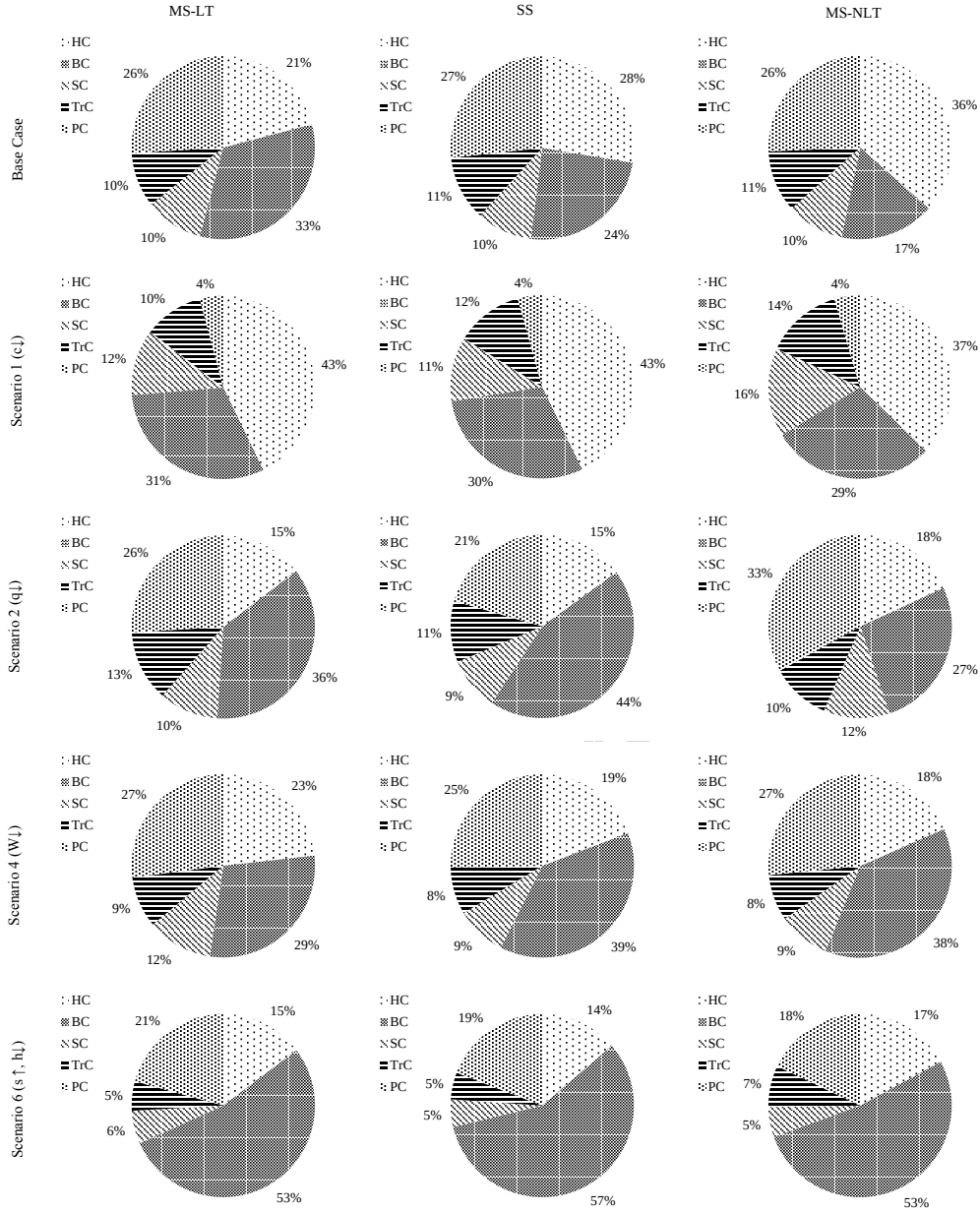


Figure 8: Operational cost contributors in select parameter scenarios.

Figure 8 shows the contribution percentage of each operational cost factor in the total operational cost within each network structure for select scenarios in parameter sensitivity experiments. As expected, when comparing scenario 1 to base case, we see a decrease in the contribution of the procurement cost in the total operational cost while giving a higher contribution to holding and backorder costs. In the MS-LT network, holding cost is a major cost contributor, while in the MS-NLT network, backorder cost has an increased contribution. On the other hand, in the SS network,

holding and backorder costs increase simultaneously. In scenario 2, where supplier qualities drop, according to the setting in Table 5, we observe increased backorder cost and reduced holding cost. As expected, with lower quality offered by suppliers, the expected yield from a supplier is reduced. When supplier capacities drop, in scenario 4, SS and MS-NLT networks experience increases in the backorder cost contribution and decreases in the holding cost. In scenario 6, we observed a decrease in all operational cost components while the contribution of the backorder costs has increased considerably.

5.3.1. Network Robustness Experiments

An important aspect of a supply chain network structure is its robustness towards the changes in the parameters of the network. In our parameter sensitivity experiments in Section 5.3, we investigated the effect of possible parameter changes on the total cost savings across all three network structures MS-LT, MS-NLT, and SS. In this section, we investigate the robustness of each one of these three network structures towards the changes in the network parameters. Specifically, with some of the changes, which we call *undesirable* changes (such as scenario 2 and scenario 4), the total costs of all three network settings increase and with some other changes, which we call *desirable* changes (such as scenario 1 and scenario 3), the total costs of all three network settings decrease when compared to the base case. However, the more interesting investigation is that which network setting has the least increase in total cost with undesirable changes and the most reduction in total cost with desirable changes in parameters. Figure 9 shows the percentage of change in total cost for each network structure within the parenthesis for each scenario next to the scenario name. Specifically, the numbers inside the parenthesis are calculated as $(\frac{C_{MS-LT}^{Scenario} - C_{MS-LT}^{BC}}{C_{MS-LT}^{BC}} \times 100, \frac{C_{MS-NLT}^{Scenario} - C_{MS-NLT}^{BC}}{C_{MS-NLT}^{BC}} \times 100, \frac{C_{SS}^{Scenario} - C_{SS}^{BC}}{C_{SS}^{BC}} \times 100)$ for each scenario where for instance C_{MS-LT}^{BC} stands for the total cost of the MS-LT network under the base case scenario. Additionally, a positive (negative) number in the parenthesis means an increase (decrease) in total cost. We observe that the MS-LT network setting has the least increase in total cost with an undesirable parameter change (see, for instance, quality decrease in scenario 2). Furthermore, when a desirable change (such as unit cost decrease in scenario 1) takes place in the network, the MS-LT network has the highest level of decrease in the total cost. We explain this due to the flexibility of the MS-LT network towards the supply chain network changes.

In case of scenario 4, where the capacities decrease, there is not a significant difference between the total cost increases. In scenarios 5 and 6, since two parameters are changing at the same time,

1 it is harder to differentiate network designs for flexibility.

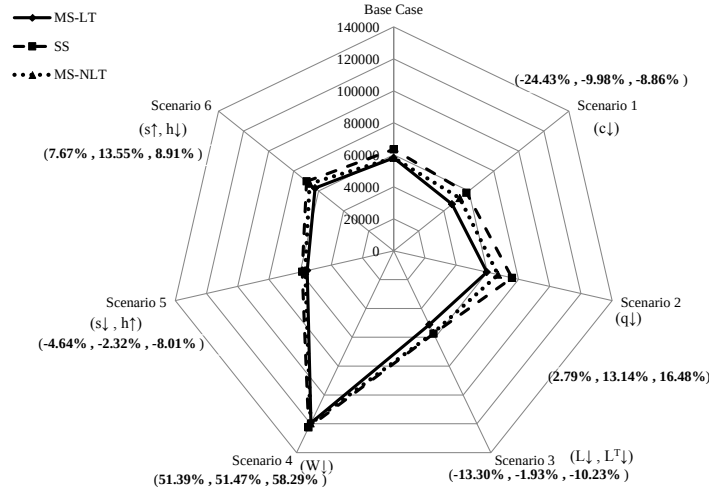


Figure 9: Robustness of different network structures.

2 5.4. Computational Time

3 An important factor in comparison of the performance of the test cases is the solution time.
4 Figure 10(a) shows this comparison among network structures under disruption test cases. The
5 SS network performs as the worst in terms of total computation time. This result was expected
6 due to the requirement of solving an MIP *TrSubProblem* by addition of the binary assignment
7 decision variables $\mathbf{A}$ and $\mathbf{A}^T$ to prevent the possibility of multi-sourcing. The MS-LT and MS-NLT
8 networks perform closely in terms of total computation time with a maximum of 229 seconds for
9 MS-LT and 253 seconds for MS-NLT network. In all of the test cases, our maximum computation
10 time does not exceed 772 seconds. The total computation time of each one of the test cases in
11 parameter sensitivity experiments are presented in Figure 10(b). As in Figure 10(a), SS network
12 takes longer to run due to the MIP *TrSubProblem* that needs to be solved. Therefore, both cost
13 and computational duration performances for MS-LT and MS-NLT dominate those of SS.

14 5.5. Service Level Experiments

15 In all of the experiments so far, we use a Type I service level (α) of 0.99 for calculating $\mathbf{R}$. In
16 this section, we present the results of experimenting with α over a range from 0.5 to 0.99 with
17 increments of 0.1. We select base case and test case 1 from the supplier disruption experiments

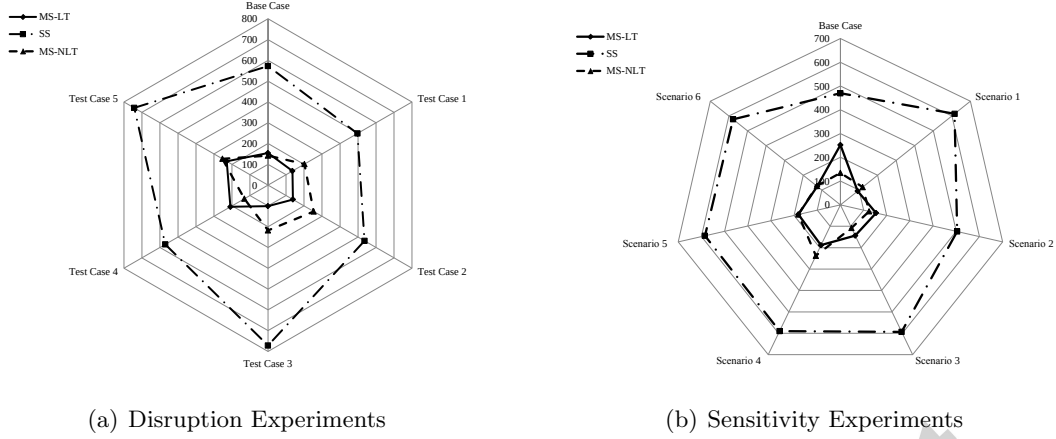


Figure 10: Computation time in experiments (given in seconds).

as the basis for experiments and we examine the results of each test case for all of our network structures. Figure 11(a) and (b) show the changes in the overall cost with respect to α for base case and test case 1, respectively.

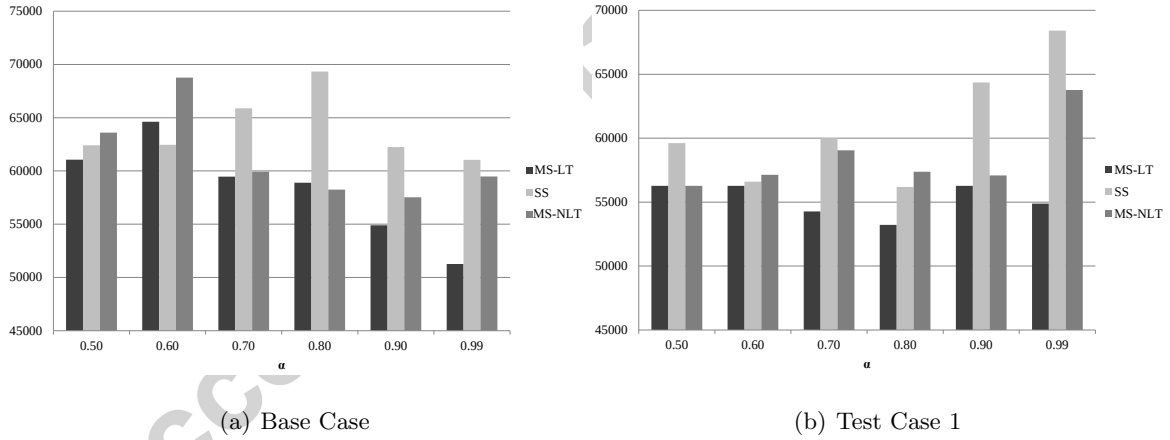


Figure 11: Total cost for different α levels.

Both in Figure 11(a) and (b), varying α impacts the total cost in a nonlinear pattern. For the short and frequent disruptions (i.e., base case), we observe that the best α values for both SS and MS-LT is at 0.99 whereas for MS-NLT, it is at 0.90. On the other hand, for the rare and long disruptions, the best α values are much lower: 0.80, 0.80, and 0.50 for MS-LT, SS, and MS-NLT, respectively. A lower α value dictates a lower $\mathbf{R}$, and correspondingly, a lower safety stock. These results from Figure 11(a) show that it is better to set higher service levels for a better overall cost performance when experiencing short and frequent disruptions. On the other hand, the same level

of service would result in higher total cost under rare but long disruptions, as observed in Figure 11(b).

We explain these observations by further examining the trade-off between backorder and holding costs. Figure 12(a) and (b) show the behavior of the holding and backorder costs with respect to changes in α for networks MS-LT, SS, and MS-NLT for the base case (frequent and short disruptions) and test case 1 (rare and long disruptions), respectively. As α is increasing, the inventory holding costs are increasing at a decreasing rate and eventually leveling for all of the network structures for both base case and test case 1. On the other hand, for backorder costs there is a striking difference between base case and test case 1. In base case, as α levels increase, the backorder costs decrease. Hence, together with the holding cost trend, it turns out that the higher service levels are most efficient. On the other hand, in test case 1, as α level increases, backorder costs show less sensitivity towards the α level. Therefore, in test case 1 with rare and long disruptions, there is not such an obvious trade-off resolution as in the base case with short and frequent disruptions. Both of these findings support the total cost analysis for different α .

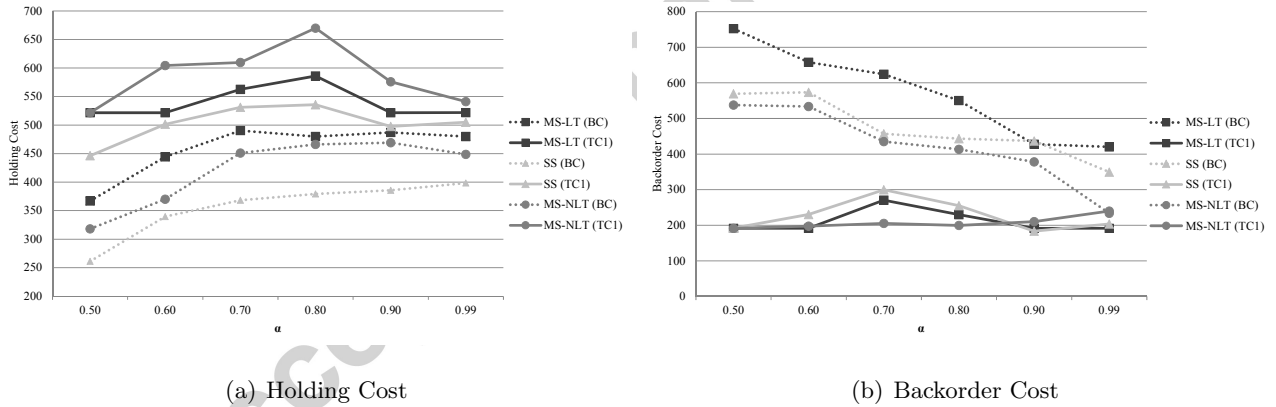


Figure 12: Holding and backorder costs for different α levels.

5.6. Supplier Selection Analysis

As a final analysis, we investigate the important factors that play pivotal roles in selection of suppliers. As evidenced by recent literature in [36] and demonstrated in Section ??, certain parameters are more influential than others. In order to understand their impact, we mark the factors that affected the supplier selection. These factors range from single influencing factors as contractual cost (f) and capacity (W) to two way interaction of these factors as $\frac{f}{W}$ and $\frac{v}{\theta}$. Notice that the interaction between the factors can have many combinations, however the most meaningful combi-

1 nations are presented here. For example, $\frac{f}{W}$ defines the amount of contractual cost charged by each
 2 supplier per each unit of capacity offered by that supplier and $\frac{\nu}{\theta}$ appears in the long run percentage
 3 of time that a supplier is available as $\frac{\nu}{1+\nu}$. Based on frequency of selection of the suppliers within
 4 each experiment type and test case, we realized the important factors as Table 6.

5 In Table 6, we first list the single factors, followed by important two way interactions and others
 6 for all network types (i.e., MS-LT, SS, and MS-NLT networks).

Table 6: Factors influencing supplier selection.

Experiment	Test Case	f	c	q	W	$\frac{\nu}{\theta}$	$\frac{f}{W}$	$f * c$	$W * q$	Other
	Base Case	✓	✓	✓						
Supplier Disruption	Test Case 1									✓
	Test Case 2					✓	✓			
	Test Case 3	✓				✓	✓			
	Test Case 4	✓			✓		✓			✓
	Test Case 5	✓			✓	✓	✓			
Parameter Sensitivity	Scenario 1	✓		✓	✓					
	Scenario 2	✓	✓		✓	✓	✓			
	Scenario 3	✓				✓	✓			
	Scenario 4				✓	✓		✓		✓
	Scenario 5	✓			✓	✓	✓			✓
	Scenario 6							✓	✓	✓

7 Table 6 suggests that, in most of the cases, the selection of the suppliers is explained by a limited
 8 number of factors. More specifically, only in test case 1, we have a more complicated scheme
 9 for selecting suppliers. In other test cases and scenarios, the selections are mainly based on one
 10 and/or two way interactions of the factors. Surprisingly, unit cost is not selected as the most
 11 important factor as frequently as others. Specifically, in test cases 2 and 3, the interaction of
 12 the contractual cost with capacity and disruption characteristics of suppliers are the key factors
 13 influencing the selection of suppliers. However, the contractual costs or supplier capacity alone are
 14 not as influential individually, especially in test case 2.

15 On the other hand, with parameter experiments, in scenario 1, contractual cost remains important
 16 along with disruption characteristics and quality. That is, when unit costs are decreased in a
 17 supply chain network, the reliable suppliers with better contractual cost and quality levels are more

desirable. In scenarios 2, 3, 4, and 5, capacity, disruption characteristics, and their interaction help determine the selection of suppliers. However, in scenario 6, where backorder costs are increased and holding costs are decreased, the interaction of quality and capacity becomes important in supplier selection.

Overall, for both supplier disruption experiments and parameter sensitivity scenarios, contractual cost (f), disruption characteristics ($\frac{v}{\theta}$), and contractual cost per unit of capacity ($\frac{f}{W}$) are the most influential factors, as opposed to unit cost identified in the literature.

6. Conclusions

In this paper, with considerations for multi-sourcing and lateral transshipment, we develop an integrated supplier selection and inventory model using optimization via simulation. The optimization component decomposes the problem into two subroutines within a hybrid heuristic. The simulation component with its five modules guides the optimization component while sorting the dynamic events and disruptions. The overall approach determines the most effective strategic (supplier selection) and tactical (inventory management) decisions with multi-sourcing and proactive lateral transshipments while considering disruptions and other complicating constraints.

From the analysis, contractual costs, capacity, and disruption characteristics of suppliers, and interactions of these factors are found to be the most significant factors in supplier selection. Managers for suppliers can increase their appeal in the selection process by reducing their contractual cost while increasing their capacity and resiliency towards disruptions. Managers for manufacturing firms can reduce their operating costs by selecting suppliers with the lowest contractual cost, highest capacity, and most favourable disruption characteristics. In addition, transshipments within the manufacturing firm help to reduce cost, especially when contractual costs are high relative to backorder costs and transportation costs. Overall, transshipments appear to shift transportation cost control from the suppliers to the warehouses. Therefore, a decision to use transshipments must be analysed almost as thoroughly as the supplier selection problem. We observed that savings obtained from multi-sourcing and lateral transshipments can be as big as 18.01% and 12.59%, respectively. Frequent and short disruptions are less costly compared to rare and long disruptions. Supply chain managers should take steps in collaborating with their suppliers in order to make their disruptions as short of a period as possible with possible contracts on the maintenance activities. When a supply chain is restricted with rare and long disruptions, it is especially useful to take

advantage of the MS-LT network setting. Regardless of the structure of the supply chain, it is especially important to work with the suppliers who have higher capacity. In other words, higher capacity is one of the main drivers even if the supply chain needs to resort to a single sourcing setting. The MS-LT network setting not only results in significantly lower total costs, but it is also a lot more robust towards the undesirable changes in the parameters of the network when compared to an SS or MS-NLT network. MS-LT setting captures more gains, compared to counterparts, from reductions in total cost when a desirable change occurs in the network such as decrease in unit cost or lead times. As opposed to the intuition, we also show that increasing the service level in frequent and short disruption settings can achieve lower overall cost.

In the paper, we noted two possible future work that stems from this research. To reiterate, one immediate research idea is related to modeling and handling of backorders. It would be interesting to eliminate back orders by incorporating emergency lateral transshipments with proactive lateral transshipments. Investigation of the trade off among lateral transshipments and backorders is quite scarce in literature. Another immediate research idea is related to solving the inventory subproblem via different approaches.

Additional future work may involve including additional costs in the total relevant costs in order to gain more insights into the value of strategic supply chain decision making. One such example is quantity discounts related to unit, contractual, or transportation costs. Another such example is to explicitly incorporate maintenance costs to eliminate or avoid disruptions. Both of these cases cause shifts in the supply chain dynamics that would require new analytical tools. Another interesting case would be related to global supply chains and considering extensions associated with exchange rates, customs, and tariffs. The models presented in this paper would be a good starting point for these extensions.

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